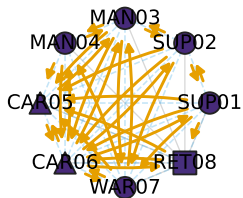
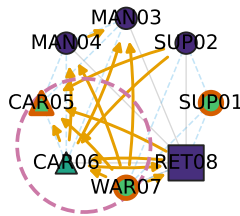


Commercial coordination network through disruption and recovery

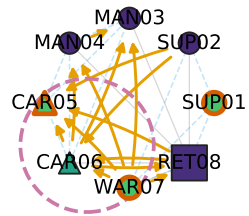
Nominal
period 0



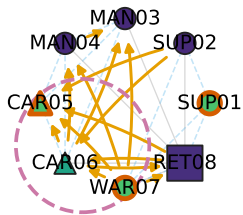
Shock onset
period 6



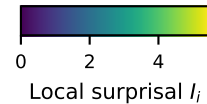
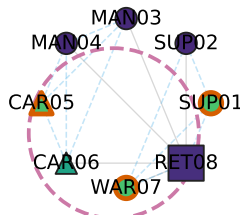
Negotiation escalation
period 6



Coalition response
period 6



Recovery endpoint
period 19



Network encoding

- Physical route
 - Recent material flow
 - Communication
 - Recent negotiation
 - Accepted commitment
 - Coalition
- Source
 - Demand
 - Transport
 - Low pressure/load
 - High pressure/load